

Theoretical Exercises – Sheet 2

- Recap the values for the attribute *age* from Exercise Sheet 2. The data tuples were 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70 (in increasing order).
 - Use smoothing by bin means (use a bin of depth 3)
 - What is the effect of smoothing for the given data?
 - What other methods are there for smoothing?
 - How might you determine outliers in the data?
- Suppose that a hospital tested the age and the glucose level data for 6 randomly selected adults as follows:

<i>age</i>	43	21	25	42	57	59
<i>glucose</i>	99	65	79	75	87	81

- Draw a scatter plot based on these two attributes.
 - Calculate the correlation coefficient by hand and interpret the result.
- Given is the following data that shows the observed number of fines in relation to the size of the car:

	<i>Carsize</i>	
<i>NrOffFines</i>	Compact	Luxury
1 or less	30	0
2 or 3	20	10
more than 3	10	30

Excerpt from a chi-square distribution table:

DF\AREA	.995	.990	.975	.950	.900	.750	.500	.250	.100	.050
1	0.00004	0.00016	0.00098	0.00393	0.01579	0.10153	0.45494	1.32330	2.70554	3.841
2	0.01003	0.02010	0.05064	0.10259	0.21072	0.57536	1.38629	2.77259	4.60517	5.991
3	0.07172	0.11483	0.21580	0.35185	0.58437	1.21253	2.36597	4.10834	6.25139	7.814

- Calculate the marginal distributions and the expected values if *Carsize* and *NoOffFines* would be independent.
- Check if *Carsize* and *NrOffFines* depend on each other (use $DOF = 2$, $\alpha=0.1$)

4. Shortly discuss issues to consider when merging data from different sources.
5. What are the value ranges of the following normalization methods?
 - a. min-max normalization
 - b. normalization by decimal scaling
 - c. z-score normalization
6. Remember the data for the attribute *age*. The *age* values for the data tuples are:
 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70
 (in increasing order).
 - a. Use min-max normalization to transform the value 35 for *age* onto the range [0,1] and [-1,1]
 - b. Use normalization by decimal scaling to transform the value of 35.
 - c. Use z-score normalization to transform the value of 35, where the standard deviation of *age* is 12.94
 - d. Comment on which normalization method you would prefer to use for the given data and why.
7. Why do we need normalization what is the difference in normalization instead of just keeping the data in its base format?
8. Theme park visitors have been recorded as by various criteria including their *age*:

<i>No</i>	1	2	3	4	5	6	7	8	9	10	11	12
<i>Age</i>	7	22	9	12	22	36	11	13	38	64	7	8

- a. Do an equidepth binning on the attribute *Age*. When done the attribute should hold just 3 distinct values. Decide on meaningful labels for the created bins. Always specify the bins' lower and upper boundaries.
- b. Now do an equiwidth binning on the attribute *Age* – again with 3 bins. What would be meaningful labels for the bins this time? Again specify the bins' lower and upper boundaries.
- c. Which discrete value for the attribute "Age" will the following unknown instances get assigned?

Age	equidepth	equiwidth
8 years		
10 years		
22 years		
45 years		
6 years		
70 years		

9. Use the data from exercise 3 and sketch examples of each of the following sample techniques: SRSWOR, SRSWR and stratified sampling. Use a sample size of 5 and the strata "youth", "middle-aged" and "senior".